

Laxmi Gummadavelly

Data Engineer

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SUMMARY

Detail-oriented Data Engineer with 5+ years of experience in designing, building, and optimizing large-scale data pipelines and cloud-native architectures across AWS, Azure, and GCP ecosystems. Skilled in Python, SQL, Spark, Kafka, Snowflake, and Airflow, with proven success in transforming raw data into scalable, high-quality analytics solutions that improve business decision-making. Adept at ETL/ELT development, data modeling, performance tuning, automation, and real-time streaming pipelines, delivering 40% faster data availability and 30% cost savings. Strong collaborator in Agile environments, partnering with cross-functional teams to support enterprise data strategies.

PROFESSIONAL EXPERIENCE

Data Engineer | Cisco Systems | USA

July 2024 – Current

- Designed and deployed scalable ETL pipelines using PySpark, Airflow, and AWS Glue, automating complex ingestion, transformation, and scheduling workflows that reduced manual data processing by 45% while enabling near real-time data availability across enterprise BI and analytics platforms.
- Built and optimized enterprise-scale data lake architectures on AWS (S3, Redshift, EMR, Lambda), supporting 20+ business units with centralized, governed data access that improved query response times by 40% and enabled advanced self-service analytics capabilities.
- Implemented real-time streaming pipelines using Kafka and Kinesis to process 1M+ events per minute, ensuring sub-200ms latency for IoT and customer usage data across global systems, which improved system responsiveness, scalability, and real-time decision-making.
- Developed Snowflake data warehouse solutions with optimized partitioning, clustering, and caching strategies, streamlining query performance and improving report generation efficiency by 30% for executive dashboards and enterprise BI reporting.
- Engineered dimensional data models and schemas (Star, Snowflake) in Redshift and Snowflake, applying indexing, partitioning, and clustering strategies that improved query performance and reduced storage costs by 20%, while enabling scalable analytics for enterprise users.
- Automated CI/CD pipelines for data workflows using Git, Jenkins, and Docker, streamlining build, test, and deployment processes that cut release times by 35% while ensuring robust version control, reliability, and traceability for all data pipeline changes.
- Partnered with data scientists and analysts to integrate machine learning pipelines (Spark MLlib, TensorFlow) into production, operationalizing models that enabled churn prediction with 93% accuracy and supported scalable deployment for advanced analytics use cases.
- Developed data validation and governance frameworks using Python, SQL, and AWS Lake Formation, implementing automated quality checks and access controls that ensured GDPR/HIPAA compliance, strengthened audit readiness, and improved overall data integrity by 25%.
- Collaborated with cross-functional teams under Agile/Scrum methodology, translating business requirements into scalable data solutions that enhanced financial forecasting accuracy, delivered operational insights, and ensured alignment with strategic objectives.

Data Engineer | Hexaware Technologies | India

June 2018 – July 2022

- Built and maintained large-scale ETL processes using Python, SQL, and Talend, integrating structured and unstructured data from ERP, CRM, and third-party APIs into centralized data warehouses, enabling unified reporting, advanced analytics, and faster decision-making.
- Designed and implemented Azure Data Factory pipelines to orchestrate batch and real-time data flows, optimizing scheduling and parallelization to reduce data refresh cycles from 8 hours to 1 hour, significantly improving decision-making speed and agility.
- Migrated legacy on-premise SQL Server workloads to Snowflake and Azure Synapse, modernizing data infrastructure to cut costs by 28% while improving scalability, query efficiency, and support for advanced analytics across business units.
- Implemented Kafka-based streaming data pipelines to capture and process real-time transactions, enabling operational visibility, enhancing fraud detection capabilities, and ensuring low-latency data delivery for financial clients.
- Optimized SQL queries, stored procedures, and indexing strategies, improving data retrieval times by 35% across enterprise reporting systems.
- Created interactive data visualization dashboards in Power BI and Tableau using DAX measures and complex SQL joins, enabling executives to monitor KPIs, track supply chain performance, analyze revenue trends, and make faster, data-driven strategic decisions.
- Designed data lake solutions on Azure (Data Lake Gen2, Synapse) integrating IoT, social media, and transactional datasets for predictive analytics use cases.
- Supported machine learning model deployment by building feature engineering pipelines, automating retraining workflows, and enabling API-based model consumption, which improved scalability, reduced manual intervention, and accelerated adoption of predictive insights.
- Implemented data governance frameworks with metadata management, lineage tracking, and access control policies, strengthening audit readiness and ensuring compliance with ISO and SOC2 requirements while improving transparency and security across enterprise data assets.
- Collaborated with global stakeholders and business analysts in Agile projects, gathering requirements, aligning priorities, and ensuring delivery of data solutions that met business SLAs while reducing time-to-market for analytics initiatives.

TECHNICAL SKILLS

- Programming & Scripting:** Python (Pandas, NumPy, PySpark), SQL (T-SQL, PL/SQL), R, Scala, and Shell scripting for data engineering and analytics workflows.
- Big Data & Processing:** Apache Spark, Hadoop, MapReduce, Hive, Pig, Spark SQL, and distributed computing for large-scale data pipelines and analytics.
- ETL & Workflow Orchestration:** Apache Airflow, Talend, Informatica, AWS Glue, Azure Data Factory, dbt (Data Build Tool) for batch and real-time data flows.
- Databases (SQL & NoSQL):** SQL Server, Oracle, PostgreSQL, MySQL, MongoDB, Cassandra, DynamoDB, and Redis for structured and unstructured data storage.
- Cloud Platforms & Services:** AWS (S3, Redshift, EMR, Lambda, Kinesis, Glue), Azure (Data Lake, Synapse, Databricks), GCP (Big Query, Pub/Sub, Dataflow) for end-to-end cloud data solutions.
- Data Warehousing:** Snowflake, Amazon Redshift, Teradata, Azure Synapse, and Google Big Query for enterprise analytics and OLAP workloads.
- Streaming & Messaging Systems:** Apache Kafka, AWS Kinesis, RabbitMQ, Spark Streaming, Flink, and Pub/Sub for high-throughput, low-latency real-time data pipelines.
- Data Modeling & Optimization:** Dimensional modeling (Star/Snowflake schemas), normalization, indexing, partitioning, clustering, materialized views, and query performance tuning.
- DevOps & CI/CD for Data:** Git, Jenkins, Docker, Kubernetes, Terraform, Ansible, and CI/CD pipelines for automated deployment of data pipelines and infrastructure-as-code.
- Visualization & Reporting Tools:** Tableau, Power BI, Looker, Superset, and Excel (Pivot, VBA, Macros) for dashboarding and KPI reporting.
- Data Governance, Quality & Security:** GDPR, HIPAA, CCPA compliance; data lineage, metadata management, role-based access control, encryption, auditing, and data masking.
- Machine Learning & Advanced Analytics:** Scikit-learn, TensorFlow, PyTorch, Spark MLlib, SageMaker, Databricks ML pipelines, feature engineering, and automated model deployment

Education

Masters of Science in Computer Science | Oklahoma City University, Oklahoma, USA

Bachelors | Sphoorthy Engineering Collage (JNTU), Hyderabad, India

Aug 2022 – May 2024

June 2014 – May 2018